

# Project Phoenix — Cloud Architecture, Multi-Region Specifications & Sprint Plan

Synthetic enterprise project document for Nanvi AI Enterprise Assistant RAG testing

## 1. Project Overview

**Project Name:** Project Phoenix

**Project Code:** PHX-2026

**Business Objective:** Build a highly available enterprise transaction and analytics platform capable of serving customers across multiple geographic regions.

**Project Sponsor:** Chief Technology Officer

**Engineering Owner:** Platform Engineering

**Target Release:** 30 November 2026

**Current Phase:** Implementation

## 2. Cloud Architecture

**Architecture Pattern:** Phoenix uses a multi-region active-active architecture. Application services are deployed independently in each production region, while globally replicated data services provide regional resilience.

**Frontend:** React and TypeScript web application delivered through a global CDN and edge routing layer.

**API Layer:** Python FastAPI services behind a managed API gateway and regional application load balancers.

**AI Services:** LangGraph-based orchestration services handle retrieval, classification, workflow execution, and AI-assisted operations.

**Primary Database:** PostgreSQL with regional read replicas and controlled cross-region replication for disaster recovery.

**Cache:** Redis is used for short-lived sessions, rate limiting, and frequently accessed metadata.

**Object Storage:** Encrypted object storage is used for documents, exports, audit artifacts, and generated reports.

**Message Processing:** A managed message queue handles asynchronous jobs such as document ingestion, report generation, notifications, and indexing.

**Observability:** Centralized logs, metrics, distributed traces, uptime checks, and alerting are required across all production regions.

## 3. Production Regions

**Region A — India:** Chennai region is the primary India-facing production region. It serves customers in India and nearby South Asian markets.

**Region B — Singapore:** Singapore provides a second production region for Asia-Pacific traffic and regional failover.

**Region C — Europe:** Frankfurt provides European service capacity and supports data residency requirements for approved European workloads.

**Traffic Routing:** Global DNS and health-based routing direct users to the nearest healthy region. Failed regional health checks cause traffic to be shifted to an available region.

**Availability Target:** Production services target 99.9% monthly availability, excluding approved maintenance windows and documented third-party outages.

## 4. Multi-Region Data Specifications

**Database Replication:** Transactional data is replicated between approved regions using managed PostgreSQL replication mechanisms. Replication lag must be monitored continuously.

**Read Strategy:** Read-heavy workloads may use regional read replicas to reduce latency.

**Write Strategy:** Critical transactional writes are routed to the designated regional writer for the tenant or workload to avoid conflicting concurrent writes.

**Consistency:** Financial and transactional operations require strong consistency at the authoritative write location. Analytics workloads may tolerate eventual consistency.

**Object Storage:** Critical documents and generated reports must have cross-region replication enabled.

**Recovery Point Objective:** Target RPO is 15 minutes for critical application data.

**Recovery Time Objective:** Target RTO is 60 minutes for critical production services.

## 5. Security Architecture

**Authentication:** Enterprise users authenticate through OIDC/OAuth2-compatible identity providers. JWT tokens are validated by backend services.

**Authorization:** RBAC is enforced for CEO, Finance, HR, Manager, Employee, and IT Admin roles. Additional attribute-based restrictions may be applied to sensitive resources.

**Secrets:** Database credentials, API keys, signing secrets, and provider tokens must be stored in an approved secrets-management service.

**Encryption:** Data must be encrypted in transit using TLS and encrypted at rest using provider-managed or customer-approved keys.

**Audit Logging:** Authentication events, privileged actions, permission changes, data exports, and administrative operations must be auditable.

## 6. Disaster Recovery

**Regional Failure:** If a production region becomes unavailable, traffic is redirected to a healthy region after automated health checks and incident procedures are initiated.

**Database Failure:** Database recovery uses the latest validated replica or backup according to the incident runbook.

**Backup Frequency:** Critical databases are backed up at least daily, with point-in-time recovery enabled where supported.

**Recovery Testing:** Disaster-recovery procedures must be tested at least once per quarter.

**Post-Incident Review:** A documented incident review is required after major outages, including timeline, impact, root-cause analysis, and corrective actions.

## 7. Sprint Milestones

**Sprint 1 — Architecture Foundation:** Finalize system architecture, service boundaries, cloud accounts, repository structure, CI/CD baseline, and environment configuration.

**Sprint 2 — Authentication & RBAC:** Implement OIDC login, JWT validation, role mapping, authorization middleware, and initial audit events.

**Sprint 3 — Core API Services:** Build FastAPI service skeletons, API gateway integration, health checks, request validation, and error handling.

**Sprint 4 — Database & Data Layer:** Provision PostgreSQL, migrations, connection pooling, replication configuration, backup policies, and data-access layer.

**Sprint 5 — Document & RAG Pipeline:** Implement document ingestion, chunking, embeddings, vector retrieval, metadata filtering, citations, and retrieval evaluation.

**Sprint 6 — Multi-Region Deployment:** Deploy application services to Chennai, Singapore, and Frankfurt; configure global routing and regional health checks.

**Sprint 7 — Observability & Security:** Add centralized monitoring, traces, security alerts, audit logging, secret management, and vulnerability checks.

**Sprint 8 — Resilience Testing:** Run failover tests, database recovery tests, load tests, chaos scenarios, and RPO/RTO validation.

**Sprint 9 — UAT & Production Readiness:** Complete user acceptance testing, documentation, runbooks, security review, performance tuning, and release approval.

## 8. Current Sprint Status

**Current Sprint:** Sprint 6 — Multi-Region Deployment

**Completed:** Core FastAPI services, authentication foundation, PostgreSQL schema, initial RAG pipeline, and CI/CD baseline.

**In Progress:** Regional deployments, global traffic routing, cross-region object replication, and health-check automation.

**Blocked:** Production cloud approval for the Frankfurt environment is pending.

**Next Milestone:** Complete multi-region smoke testing and begin resilience validation.

## 9. Key Technical Requirements

**Latency:** Target API p95 latency is below 500 ms for standard non-AI requests under normal load.

**AI Response:** RAG responses should return source references and avoid presenting unsupported information as verified company facts.

**Scalability:** Services must support horizontal scaling without relying on local filesystem state.

**Stateless Services:** Application containers should remain stateless; persistent state must be stored in approved databases, caches, or object storage.

**Health Checks:** Every production service must expose liveness and readiness checks.

**Deployment:** Production deployments must use CI/CD with approval gates and rollback capability.

## 10. Risks & Mitigations

**Cross-Region Data Conflicts:** Use authoritative write ownership and explicit consistency rules for transactional workloads.

**Replication Lag:** Monitor lag and prevent dependent workflows from reading stale data where strong consistency is required.

**Regional Outage:** Maintain active capacity in multiple regions and test automated/manual failover procedures.

**Cloud Cost Growth:** Use budgets, resource tagging, autoscaling policies, and monthly cost reviews.

**AI Hallucination:** Require retrieval grounding, source citations, confidence checks, and fallback responses when evidence is unavailable.

## 11. Frequently Asked Questions

**Q: What is Project Phoenix?:** A: Project Phoenix is a multi-region enterprise platform designed for high availability, regional resilience, secure data processing, and AI-assisted operations.

**Q: Which regions are planned?:** A: Chennai, Singapore, and Frankfurt.

**Q: Is Phoenix active-active?:** A: Yes. The application tier is designed for multi-region active-active operation, while transactional write ownership follows defined consistency rules.

**Q: What is the target RPO?:** A: 15 minutes for critical application data.

**Q: What is the target RTO?:** A: 60 minutes for critical production services.

**Q: What sprint handles multi-region deployment?:** A: Sprint 6.

**Q: What is blocking the Frankfurt deployment?:** A: Production cloud approval for the Frankfurt environment.

**Q: What is the API latency target?:** A: p95 below 500 ms for standard non-AI requests under normal load.

**Q: What happens if a region fails?:** A: Global health-based routing redirects traffic to an available healthy region, followed by incident and recovery procedures.

**Q: Which authentication standard is used?:** A: OIDC/OAuth2-compatible enterprise authentication with backend JWT validation.

## 12. RAG Search Metadata

**Keywords:** Project Phoenix, PHX-2026, cloud architecture, multi-region, Chennai, Singapore, Frankfurt, active-active, PostgreSQL, Redis, FastAPI, LangGraph, RAG, disaster recovery, RPO, RTO, Sprint 6, Sprint 7, Sprint 8, failover, architecture, cloud deployment.

**Departments:** Platform Engineering, AI Engineering, DevOps, Security, IT, Product.

**Document Type:** Technical Architecture / Project Plan / Engineering Specification

**Year:** 2026

*MOCK DATA NOTICE: Synthetic data for RAG retrieval and chatbot testing. It is not a real cloud architecture, production specification, or company commitment.*